

GARTLAND, Norway

WMO: 012910

Lat: 64.531N

Lon: 12.384E

Elev: 312

StdP: 14.53

Time Zone: 1.00 (EUC)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.0	(c) -2.5	(d) -13.2	(e) 2.7	(f) -8.5	(g) -6.7	(h) 3.9	(i) -1.9	(j) 23.5	(k) 24.5	(l) 21.0	(m) 25.1	(n) 2.0	(o) 240	(p) 0.644

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.2	79.4	63.6	74.8	61.4	71.1	59.4	65.4	75.9	62.9	72.2	60.8	69.1	5.8	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.2	81.5	69.3	59.0	75.5	66.6	57.0	70.2	63.6	30.4	75.8	28.6	72.1	27.1	69.5	74.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.9	14.3	12.1	-11.5	83.6	8.2	5.1	-17.3	87.3	-22.1	90.3	-26.7	93.1	-32.6	96.9
			-11.6	88.2	7.9	3.2	-17.3	70.5	-21.9	72.4	-26.3	74.2	-32.1	76.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.8	22.6	23.4	28.7	36.8	45.5	52.1	58.1	55.1	49.0	38.7	29.7	24.8		
	DBStd	14.80	11.08	10.59	8.69	5.43	7.25	6.42	6.38	4.81	5.34	6.21	9.70	10.96		
	HDD50	4715	849	746	659	395	181	45	5	8	80	355	610	782		
	HDD65	9586	1314	1166	1124	845	604	390	231	308	481	816	1060	1247		
	CDD50	628	0	0	0	0	43	109	256	167	49	4	0	0		
	CDD65	23	0	0	0	0	0	4	17	2	0	0	0	0		
	CDH74	467	0	0	0	0	30	82	305	50	0	0	0	0		
	CDH80	103	0	0	0	0	2	14	85	2	0	0	0	0		
	WSAvg	4.6	5.4	5.3	5.2	4.5	4.9	4.5	4.1	3.9	4.1	4.4	4.2	4.6		
	PrecAvg	51.3	5.3	4.5	4.6	3.1	2.5	3.3	3.6	3.7	5.0	5.2	4.8	5.9		
(15) Precipitation	PrecMax	67.4	11.8	11.5	10.4	6.6	3.6	6.5	8.0	6.3	12.4	8.4	10.0	10.7		
	PrecMin	35.2	0.6	0.5	0.7	1.1	1.0	0.3	1.7	1.5	1.3	1.7	0.5	1.8		
	PrecStd	7.6	2.9	2.6	2.6	1.5	0.7	1.5	1.5	1.3	2.6	1.8	2.8	2.7		
	PrecStd	7.6	2.9	2.6	2.6	1.5	0.7	1.5	1.5	1.3	2.6	1.8	2.8	2.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.8	44.0	48.0	64.4	77.5	82.2	88.0	78.9	70.7	60.6	49.5	45.3		
		MCWB	42.1	40.4	40.2	50.4	60.0	62.7	68.1	63.6	59.4	52.9	48.2	41.9		
	2%	DB	39.9	41.6	43.5	57.6	72.3	75.1	81.7	74.9	65.9	54.5	46.2	42.3		
		MCWB	38.2	39.2	39.4	46.1	56.8	59.7	65.9	61.9	57.4	49.3	44.3	40.2		
	5%	DB	37.7	39.7	41.5	51.3	67.0	70.1	76.7	70.8	62.2	51.0	42.7	39.6		
		MCWB	35.9	37.5	38.4	42.6	54.2	57.7	63.4	60.4	55.3	47.8	40.1	37.4		
	10%	DB	34.9	37.1	39.5	46.7	60.7	65.6	72.3	67.0	59.1	47.6	40.5	37.5		
		MCWB	33.7	34.8	36.7	40.3	50.7	55.8	61.3	59.1	53.6	45.4	38.5	35.5		
	0.4%	WB	42.0	41.6	42.4	50.9	61.3	63.9	71.2	66.0	60.9	53.3	48.8	43.9		
		MCDB	42.4	42.8	43.9	63.3	76.0	77.5	85.5	73.4	66.9	58.4	49.6	44.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	38.6	39.8	40.5	46.8	58.0	61.7	67.4	63.4	58.5	50.1	44.7	40.4		
		MCDB	39.3	41.3	42.6	56.6	69.8	73.2	79.0	71.1	64.5	53.4	46.1	41.4		
	5%	WB	36.0	37.8	38.7	43.7	54.6	59.1	64.7	61.3	56.3	48.0	40.7	37.6		
		MCDB	37.3	39.4	41.2	49.4	65.4	67.7	75.1	69.0	61.2	50.6	42.2	39.2		
	10%	WB	33.8	34.7	36.9	41.2	51.6	56.6	62.0	59.5	54.1	45.8	38.6	35.6		
		MCDB	34.7	36.5	39.1	45.4	59.6	63.9	70.3	66.1	58.2	47.4	40.1	37.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.4	12.5	15.0	17.7	19.8	18.3	19.2	19.3	16.8	12.0	9.4	10.7		
		MCDBR	8.9	11.1	14.1	27.7	31.3	28.1	29.4	27.7	23.0	16.0	10.7	11.1		
		MCWBR	7.8	8.8	10.3	17.0	17.0	14.4	14.9	16.1	14.9	11.6	9.2	9.5		
		MCDBR	8.6	10.2	12.2	23.7	28.5	23.9	26.6	24.5	21.4	14.0	10.3	10.7		
	5% WB	MCWBR	7.9	8.9	9.7	15.4	16.0	12.7	14.1	15.1	14.5	10.7	9.1	9.5		
		MCWBR	7.9	8.9	9.7	15.4	16.0	12.7	14.1	15.1	14.5	10.7	9.1	9.5		
(40) Clear-Sky Solar Irradiance	taub		0.213	0.282	0.313	0.342	0.346	0.345	0.353	0.340	0.323	0.301	0.345	0.366		
	taud		1.961	2.160	2.208	2.297	2.415	2.470	2.470	2.496	2.535	2.409	2.604	2.625		
	Ebn at Noon		102	194	241	263	274	277	271	264	245	193	98	44		
	Edh at Noon		8	19	28	32	31	30	30	26	21	15	7	4		
(44) All-Sky Solar Radiation	RadAvg		38	180	577	1157	1467	1510	1444	1104	612	252	62	15		
	RadStd		4	31	95	113	102	186	176	122	70	29	9	1		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (36 neighbors)	+0.85	N/A	N/A	N/A	N/A	N/A	-256	-320	N/A	N/A

Nomenclature: See separate page